
Coin Toss

Finance and Budgeting Manager

Preliminary Project Specification

- Track income, expenses, and transfers with receipts
- Set budgets with weekly, monthly, or yearly limits
- Create savings goals and track progress over time
- View analytics with charts and health scoring
- Multi-currency support (7 currencies)
- Automated recurring transactions via daily cron job

Tech Stack: React 18 | TypeScript | Express | Prisma | PostgreSQL (Supabase) | Vercel

A. System Description

1. What the System Does

Coin Toss is a personal finance and budgeting web application. It helps users track their income, expenses, budgets, and savings goals in one place. The system provides a visual dashboard with charts, analytics, and notifications to give users a clear picture of their financial health. It replaces spreadsheets and complex budgeting software with a simple, intuitive interface accessible from both desktop and mobile devices.

2. Target Audience

People who want a clear picture of where their money goes. This includes young professionals starting to budget, freelancers tracking income from multiple sources, families managing household finances, and anyone who prefers a simple visual tool over complicated accounting software.

3. What Need It Addresses

Managing personal finances with spreadsheets is error-prone, hard to visualize, and difficult to maintain. Existing budgeting apps are often too complex or locked behind paywalls. Coin Toss solves this by providing a free, simple, visual tool that automatically tracks spending, sets budgets, monitors savings goals, and sends alerts when budgets are close to their limits.

Key Features

Sign up / Log in	JWT auth with access + refresh tokens, httpOnly cookies, account lockout after 5 failed attempts
Track transactions	Add income, expenses, and transfers. Search, filter, sort, and upload receipts
Set budgets	Recreate category limits with weekly, monthly, or yearly periods. Color-coded progress bars
Savings goals	Set targets and track progress. Add contributions with atomic increments
View analytics	Networks, monthly trends, category breakdowns, and a financial health score (0-100)
Categories	Create and delete categories. Deletion reassigns data atomically to fallback
Notifications	Alerts when budgets approach limits or goals are reached
Recurring transactions	Templates for repeating payments, auto-created by a daily cron job
Multi-currency	7 currencies with real-time exchange rates from frankfurter.app
Settings	Edit profile, change password, switch currency, toggle dark mode, delete account

B. Wireframes / Mockups

The design follows a Minimalist SaaS Dashboard style with a warm brown color scheme (#704c35). The app uses card-based layouts, chart visualizations, and a responsive design that adapts between desktop (sidebar navigation) and mobile (bottom navigation bar).

1. Dashboard View

The main landing page after login. Features a left sidebar (desktop) or bottom nav (mobile) with links: Dashboard, Transactions, Budgets, Goals, Analytics, Notifications, Settings. Content: four KPI cards (Balance, Income, Expenses, Savings), two charts (bar + pie), a budget usage progress bar, and the five most recent transactions.

```

+-----+
| Dashboard                                     Health: 72 |
+-----+
|
| +-----+ +-----+ +-----+ +-----+ |
| | Balance | | Income | | Expense | | Save | |
| | $12,450 | | $5,230 | | $3,180 | | $2,050 | |
| +-----+ +-----+ +-----+ +-----+ |
|
| +-----+ +-----+ |
| | Income vs Expense | | Income vs Expense | |
| | (Bar Chart) | | (Pie/Donut) | |
| +-----+ +-----+ |
|
| Budget Usage: 48% of $3,000 [=====] |
|
| +-- Recent Transactions -----+ |
| | Grocery Store   -$85.00   Jun 15 | |
| | Salary          +$12,000   Jun 01 | |
| | Electric Bill   -$320.00   Jun 03 | |
| +-----+ |
+-----+

```

2. Transactions Page

Searchable, filterable table (desktop) or card list (mobile) of all transactions. Search bar, type filter (All / Income / Expense / Transfer), pagination (15 per page). Each row shows description, category, date, type badge, and amount. Add Transaction button opens a dialog form.

```

+-----+
| Transactions                                [+ Add Transaction] |
+-----+
| [Search] [Type: All v] |
|
| Description | Category | Date | Type |
| Grocery Store | Food | Jun 15 | EXP |
| Salary | Work | Jun 01 | INC |
| Electric Bill | Utilities | Jun 03 | EXP |
|
| < Page 1 of 3 > |
+-----+

```

3. Budgets Page

Budget cards in a 2-column grid (desktop) or single column (mobile). Each card: category name with color dot, spent vs limit, progress bar. Bar colors: brown under 90%, yellow 90-99%, red at 100%+.

```

+-----+
| Budgets                                     [+ Add Budget] |
+-----+
|
| +-----+ +-----+ |
| | Food | | Rent | |
| | $450 / $600 | | $1,200 / $1,200 | |
| | [===== 75%] | | [===== 100%] | |
| | [Edit] [Delete] | | [Edit] [Delete] | |
| +-----+ +-----+ |
+-----+

```

4. Goals Page

Savings goal cards in a 2-column grid (desktop) or single column (mobile). Each card: goal name, current vs target, progress bar with %, optional deadline, and Contribute / Edit / Delete buttons.

+-----+-----+			
	Savings Goals		[+ Add Goal]
+-----+-----+			
	+-----+	+-----+	
	Vacation	Emergency	
	\$1,200 / \$3,000	\$800 / \$5,000	
	[===== 40%]	[== 16%]	
	Due: Sep 15		
	[Contribute]	[Contribute]	
	[Edit] [Delete]	[Edit] [Delete]	
	+-----+	+-----+	
+-----+-----+			

5. Analytics Page

Four KPI cards (average spending, savings rate, health score, net worth), area chart for monthly spending trends (6 months), pie chart for expense breakdown by category, bar chart for cash flow over 12 months. All charts stack vertically on mobile.

6. Additional Pages

Login: Centered card on a gradient background with email, password, remember me checkbox, links to register and forgot password.

Register: Similar layout with name, email, password, and confirm password fields.

Settings: Six sections — Profile, Password, Appearance (dark mode toggle), Currency and Locale, Session (sign out), and Danger Zone (delete account).

Notifications: List of notifications with unread count badge, mark all read button, and delete per notification.

C. DB Schema (ERD)

PostgreSQL (via Supabase) with 8 tables. All money amounts stored in USD using Decimal(12,2). Passwords hashed with bcrypt (cost 12). Reset tokens hashed with SHA-256.

1. users

```
{
  id:          UUID,          // Primary key, auto-generated
  name:        String,        // Display name (2-100 chars)
  email:       String,        // Login email, unique
  password_hash: String,      // Bcrypt hash (cost 12)
  role:        Enum,          // 'USER' | 'ADMIN' (default: USER)
  is_verified: Boolean,       // Email verified? (default: false)
  is_locked:   Boolean,       // Account locked? (default: false)
  failed_logins: Integer,     // Failed attempts (default: 0)
  lock_until:  Timestamp,     // Lock expiry (null if not locked)
  reset_token: String,        // SHA-256 hash of reset token (null)
  reset_expires: Timestamp,   // Reset token expiry (1 hour)
  currency:    String,        // Display currency (default: USD)
  locale:      String,        // Locale (default: en-US)
  token_version: Integer,     // Invalidate tokens (default: 0)
  created_at:  Timestamp
  updated_at:  Timestamp
}
```

2. transactions

```
{
  id:          UUID,          // Primary key
  user_id:     UUID,          // FK -> users.id (cascade)
  category_id: UUID,          // FK -> categories.id (set null)
  amount:      Decimal(12,2), // Amount in USD, always positive
  description: String,        // 1-255 chars
  type:        Enum,          // INCOME | EXPENSE | TRANSFER
  date:        Timestamp,     // When it happened
  payment_method: String,     // Optional (max 50 chars)
  notes:       String,        // Optional (max 1000 chars)
  receipt_url: String,        // Vercel Blob URL (optional)
  is_recurring: Boolean,      // Auto-created by cron? (false)
  tags:        String[],      // Max 10, each max 30 chars
  created_at:  Timestamp,
  updated_at:  Timestamp
}
```

Indexes: *[user_id, date], [user_id, type], [user_id, category_id], [user_id, date, type]*

3. categories

```
{
  id:          UUID,          // Primary key
  user_id:     UUID,          // FK -> users.id (null = default)
  name:        String,        // 1-50 chars
  icon:        String,        // Lucide icon name (default: circle)
  color:       String,        // Hex color (default: #6366f1)
  type:        Enum,          // INCOME | EXPENSE
  created_at:  Timestamp,
  updated_at:  Timestamp
}
```

Unique: *[user_id, name, type]*. System defaults (*user_id = null*) cannot be edited or deleted.

4. budgets

```
{
  id:          UUID,          // Primary key
  user_id:     UUID,          // FK -> users.id (cascade)
  category_id: UUID,          // FK -> categories.id (restrict)
  limit:        Decimal(12,2), // Max spending in USD
  period:       Enum,          // WEEKLY | MONTHLY | YEARLY
  start_date:   Timestamp,     // Period start (default: now)
  end_date:     Timestamp,     // Period end (default: end of month)
  created_at:   Timestamp,
  updated_at:   Timestamp
}
```

Unique: *[user_id, category_id, period]*. Computed: *spent = expenses in range, percentage = (spent / limit) * 100*

5. savings_goals

```
{
  id:          UUID,          // Primary key
  user_id:     UUID,          // FK -> users.id (cascade)
  name:        String,        // Goal name (1-100 chars)
  target_amount: Decimal(12,2), // Savings target in USD
  current_amount: Decimal(12,2), // Saved so far (default: 0, atomic)
  deadline:    Timestamp,     // Optional (null = no deadline)
  created_at:  Timestamp,
  updated_at:  Timestamp
}
```

Computed: *progress = (current_amount / target_amount) * 100*

6. recurring_transactions

```
{
  id:          UUID,          // Primary key
  user_id:     UUID,          // FK -> users.id (cascade)
  category_id: UUID,          // FK -> categories.id (restrict)
  amount:      Decimal(12,2), // Amount per generated transaction
  description:  String,        // Description for generated transactions
  type:         Enum,          // INCOME | EXPENSE | TRANSFER
  interval:     Enum,          // DAILY | WEEKLY | MONTHLY | YEARLY
  day_of_month: Integer,       // For MONTHLY/YEARLY (1-31)
  day_of_week:  Integer,       // For WEEKLY (0=Sun - 6=Sat)
  start_date:   Timestamp,     // Default: now
  end_date:     Timestamp,     // null = forever
  next_date:    Timestamp,     // Next due date
  is_active:    Boolean,       // Default: true
  created_at:   Timestamp,
  updated_at:   Timestamp
}
```

Index: `[next_date]` — cron job queries due templates.

7. notifications

```
{
  id:          UUID,          // Primary key
  user_id:     UUID,          // FK -> users.id (cascade)
  title:        String,        // Short title
  message:      String,        // Full message body
  type:         Enum,          // INFO | WARNING | ERROR | SUCCESS
  is_read:      Boolean,       // Read? (default: false)
  created_at:   Timestamp,
  updated_at:   Timestamp
}
```

Indexes: `[user_id]`, `[is_read]`. Pagination meta includes `unreadCount`.

8. refresh_tokens

```
{
  id:          UUID,          // Primary key
  user_id:     UUID,          // FK -> users.id (cascade)
  token:        String,        // Refresh token (unique)
  family:       String,        // Family ID (same login = same family)
  expires_at:   Timestamp,     // Token expiry
  is_revoked:   Boolean,       // Revoked? (default: false)
  created_at:   Timestamp,
  updated_at:   Timestamp
}
```

Indexes: `[user_id]`, `[token]` (unique). Reusing a revoked token revokes entire family.

Table Relationships Summary

From	To	Type	On Delete
User	Transaction	1 to many	Cascade
User	Category	1 to many	Cascade
User	Budget	1 to many	Cascade
User	SavingsGoal	1 to many	Cascade
User	RecurringTxn	1 to many	Cascade
User	Notification	1 to many	Cascade
User	RefreshToken	1 to many	Cascade
Category	Transaction	1 to many	Set Null
Category	Budget	1 to many	Restrict

Category	RecurringTxn	1 to many	Restrict
----------	--------------	-----------	----------

D. User Flow

1. Registration and Login Flow

```

User opens the app
|
v
Does the user have an access token (login session)?
|
+-- No --> Show /login page
|
|       +-- User enters email + password
|       |       +-- Correct --> Store tokens --> /dashboard
|       |       +-- Wrong --> Show error message
|       |
|       +-- "Create account" --> /register
|       |       +-- Fill name + email + password
|       |       +-- Account created --> /login
|       |
|       +-- "Forgot password?" --> /forgot-password
|       |       +-- Enter email --> Send reset link
|       |       +-- /reset-password?token=xxx
|       |       +-- Enter new password --> /login
|
+-- Yes --> Try fetching data with the token
|
+-- Token valid --> /dashboard
+-- Token expired --> Try refreshing
|       +-- Works --> /dashboard
|       +-- Fails --> /login

```

Access tokens expire after 15 minutes. Refresh tokens last 1 day (or 30 days with "Remember me"). After 5 failed logins, the account locks for 15 minutes. Password changes revoke all sessions.

2. Main App Navigation

Once logged in, the user sees the dashboard. From the sidebar (desktop) or bottom nav (mobile), they can navigate to: Transactions, Budgets, Goals, Analytics, Notifications, and Settings. Each page loads its data from the API with a loading spinner, then displays the content or an error card with a retry button.

```

+-----+
| /dashboard |
| KPIs, charts, recent |
+-----+
|
+-----+
| | | | |
v v v v v
/transac- /budgets /goals /analytics /notifications
tions Add/Edit Add/Edit View Mark read
Add/Edit Delete Add/Edit View charts Delete
Delete View % Delete charts (read-only)
Search

|
v
/settings
Edit profile / Change password
Change currency / Toggle dark mode
Sign out / Delete account

```

3. Token Refresh Flow

```

Frontend sends request with expired access token
|
v
Backend returns 401 Unauthorized
|
v
Frontend intercepts the 401 error
|
+-- Another refresh already in progress?
|     +-- Yes --> Wait in queue, retry with new token
|     +-- No  --> Send POST /auth/refresh
|
|     +-- Success --> Retry original request
|     +-- Fail  --> Logout --> /login

```

The refresh token is stored in an httpOnly cookie — JavaScript cannot read it, preventing XSS attacks from stealing tokens.

4. Adding a Transaction Flow

```

User clicks "Add Transaction"
|
v
Dialog opens with form fields:
  Amount, Description, Type, Category, Date,
  Payment Method, Notes, Tags
|
v
User fills form and clicks "Save"
|
v
Frontend validates with Zod schema
|
+-- Validation fails --> Show field errors
+-- Validation passes --> API POST /transactions
                             +-- 201 Created --> Close dialog,
                             |                    refresh list
                             +-- Error --> Show error message

```

5. Category Deletion Flow

```

User clicks "Delete" on a category
|
v
Is it a system default category?
|
+-- Yes --> Return 403 Forbidden
|
+-- No --> Start database transaction:
          1. Find or create "Uncategorized" fallback
          2. Reassign all transactions to fallback
          3. Reassign all budgets to fallback
          4. Reassign all recurring templates to fallback
          5. Delete the category
          6. Commit (all or nothing)

```

Using a database transaction means if anything fails, nothing changes — no data is left orphaned.

6. Recurring Transactions (Daily Cron Job)

```

Cron job fires at midnight (Vercel Cron)
|
v
Check X-Cron-Secret header matches?
|
+-- No --> Return 401 (reject)
|
+-- Yes --> Find all active templates where next_date <= now()
          |
          +-- None found --> Return { processed: 0 }
          +-- Found N templates --> For each:
                1. Create a new Transaction (copy fields)
                2. Advance next_date to next interval
                3. If next_date > end_date, set is_active = false
                4. Return { processed: N }

```


E. API Endpoints

Base URL: /api/v1. Auth required for most endpoints (Bearer token in Authorization header). All money amounts in USD.

Authentication

Method	Endpoint	Description	Body / Query
POST	/auth/register	Create a new account	name, email, password, passwordConfirmation
POST	/auth/login	Log in and get tokens	email, password, rememberMe?
POST	/auth/logout	Log out, revoke tokens	refreshToken? (or cookie)
POST	/auth/refresh	Get new access token	refreshToken? (or cookie)
POST	/auth/forgot-password	Request password reset email	email
POST	/auth/reset-password	Reset password with token	token, password, passwordConfirmation
GET	/auth/me	Get logged-in user profile	-
PATCH	/auth/profile	Update name and/or email	name?, email?
PATCH	/auth/me/password	Change password	currentPassword, newPassword, newPasswordConfirmation
PATCH	/auth/preferences	Update currency/locale	currency?, locale?
DELETE	/auth/account	Delete account permanently	-

Transactions

Method	Endpoint	Description	Body / Query
GET	/transactions	List transactions (paginated)	page, limit, sortBy, type, search, ...
GET	/transactions/summary	Get totals for date range	startDate?, endDate?
GET	/transactions/:id	Get one transaction	-
POST	/transactions	Create a transaction	amount, description, type, categoryId?
PUT	/transactions/:id	Update a transaction	amount?, description?, ...
DELETE	/transactions/:id	Delete a transaction	-
POST	/transactions/bulk-delete	Delete multiple	ids[]

Categories

Method	Endpoint	Description	Body / Query
GET	/categories	List all categories	-
GET	/categories/:id	Get one category	-

POST	/categories	Create a category	name, type, icon?, color?
PUT	/categories/:id	Update a category	name?, icon?, color?
DELETE	/categories/:id	Delete (reassigns data)	-

Budgets

Method	Endpoint	Description	Body / Query
GET	/budgets	List budgets (paginated)	page, limit
GET	/budgets/:id	Get one budget	-
POST	/budgets	Create a budget	categoryId, limit, period?
PUT	/budgets/:id	Update a budget	limit?, period?
DELETE	/budgets/:id	Delete a budget	-

Savings Goals

Method	Endpoint	Description	Body / Query
GET	/goals	List goals (paginated)	page, limit
GET	/goals/:id	Get one goal	-
POST	/goals	Create a goal	name, targetAmount, deadline?
PUT	/goals/:id	Update a goal	name?, targetAmount?
DELETE	/goals/:id	Delete a goal	-
POST	/goals/:id/contribute	Add money to a goal	amount

Notifications

Method	Endpoint	Description	Body / Query
GET	/notifications	List (paginated)	page, limit
PATCH	/notifications/read-all	Mark all as read	-
PATCH	/notifications/:id/read	Mark one as read	-
DELETE	/notifications/:id	Delete a notification	-

Analytics

Method	Endpoint	Description	Body / Query
GET	/analytics/dashboard	Dashboard summary + KPIs	-
GET	/analytics/overview	Combined dashboard + monthly	-
GET	/analytics/monthly-spending	Income/expense by month	months? (default: 6)
GET	/analytics/category-breakdown	Expense by category	startDate?, endDate?
GET	/analytics/cash-flow	Income vs expenses over time	months? (default: 12)
GET	/analytics/net-worth	Net worth + monthly trend	-

File Uploads

Method	Endpoint	Description	Body / Query
POST	/uploads/receipt	Upload receipt file	file (multipart/form-data)
POST	/uploads/receipt/delete	Delete a receipt file	url

Recurring Transactions

Method	Endpoint	Description	Body / Query
GET	/recurring	List all recurring templates	-
GET	/recurring/:id	Get one template	-
POST	/recurring	Create a template	amount, description, type, interval
PATCH	/recurring/:id	Update a template	amount?, isActive?
DELETE	/recurring/:id	Delete a template	-

System

Method	Endpoint	Description	Body / Query
POST	/cron/recurring	Process due recurring (cron)	X-Cron-Secret header
GET	/health	Health check (no auth)	-

Summary: 53 endpoints — 7 public (no auth), 30 require JWT, 1 cron (special auth), 6 rate-limited, 4 paginated, 23 validated with Zod.

Generated from the Coin Toss Finance Manager codebase.